Chem 352L – Friedel–Crafts Acylation of Ferrocene: Synthesis of Acetylferrocene and Diacetylferrocene

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Abstract

Ferrocene was reacted with acetyl chloride in the presence of anhydrous aluminum chloride to produce a mixture of acetylferrocene and diacetylferrocene through Friedel–Crafts acylation. The crude product was examined by thin-layer chromatography (TLC), separated by alumina column chromatography, and characterized by melting point and infrared (IR) spectroscopy. From 0.100 g of ferrocene, 0.120 g of acetylferrocene and 0.065 g of diacetylferrocene were isolated. Calculated individually from the balanced equations, the percent yields were 97.8 % for acetylferrocene and 44.8 % for diacetylferrocene. Because both products derive from the same ferrocene, however, these yields sum to about 143 % of the available ferrocene, which is impossible; the isolated solids therefore must have contained solvent or other impurities, or a mass was recorded incorrectly. The experimental melting points were 85.5 °C for acetylferrocene and 112 °C for diacetylferrocene. Both IR spectra showed the expected ketone carbonyl absorption, supporting formation of the acylated products.

1 Introduction

Friedel–Crafts acylation is an electrophilic aromatic substitution reaction in which an aromatic substrate reacts with an acyl electrophile to form an aromatic ketone. In this experiment, ferrocene was acylated using acetyl chloride and anhydrous aluminum chloride. Depending on the reaction conditions and the extent of acetylation, both monoacetylferrocene and diacetylferrocene can form.

Ferrocene contains two cyclopentadienyl rings coordinated to an iron(II) center. Each cyclopentadienyl anion contains six π electrons. Therefore, each ring satisfies Hückel's rule, $4n+2$, with $n=1$, and is aromatic. The high electron density of the rings makes ferrocene especially reactive toward electrophilic substitution.

Aluminum chloride functions as a Lewis acid. It coordinates to the carbonyl oxygen or chlorine of acetyl chloride and polarizes the carbon–chlorine bond. This interaction promotes formation of the acylium ion, CH_3CO^+, which acts as the electrophile:

$$CH_3COCl + AlCl_3 \longrightarrow CH_3CO^+ + AlCl_4^- \tag{1}$$

The acylium ion attacks an electron-rich cyclopentadienyl ring. Loss of a proton restores aromaticity and produces acetylferrocene. A second acylation can occur on the other ring to produce diacetylferrocene. Further substitution on the same ring is disfavored because the acetyl group withdraws electron density through inductive and resonance effects, deactivating that ring toward another electrophilic attack.

The reaction mixture was analyzed by TLC using silica as the stationary phase and methylene chloride as the mobile phase. Silica is polar, while ferrocene is relatively nonpolar; consequently, ferrocene interacts weakly with silica and travels farther. Acetylferrocene and diacetylferrocene contain polar ketone groups and are retained more strongly. The compounds were then separated by alumina column chromatography using solvents of increasing polarity.

2 Background and Theory

2.1 Aromaticity of Ferrocene

Each five-membered cyclopentadienyl ring in ferrocene is cyclic, approximately planar, and fully conjugated. Each ring contributes six π electrons: four electrons come from two carbon–carbon double bonds and two additional electrons come from the lone pair created when the neutral cyclopentadiene ring is deprotonated. Therefore, each ring satisfies Hückel's rule:

$$4n+2=6 \qquad (n=1). \tag{2}$$

The two aromatic cyclopentadienyl anions coordinate to an iron(II) center. Because the rings are electron-rich, ferrocene reacts readily with electrophiles in electrophilic aromatic substitution reactions.

2.2 Generation of the Acylium Ion

Anhydrous aluminum chloride is a Lewis acid. It accepts electron density from the chlorine atom or carbonyl oxygen of acetyl chloride and forms a Lewis acid–Lewis base complex. This interaction weakens the acyl chloride bond and generates the acylium ion, the electrophile used in the reaction:

$$\mathrm{CH_3COCl} + \mathrm{AlCl_3} \rightleftharpoons \mathrm{CH_3CO^+} + \mathrm{AlCl_4^-}. \tag{3}$$

The acylium ion is resonance-stabilized:

$$\mathrm{CH_3{-}C{\equiv}O^+} \rightleftharpoons \mathrm{CH_3{-}C^+{=}O}. \tag{4}$$

The electron-rich ferrocene ring attacks the acylium ion to form an arenium ion intermediate. Loss of a proton restores aromaticity and produces acetylferrocene.

2.3 Overall Reaction Equations

The monoacylation reaction can be written as:

$$\boxed{\mathrm{Fe(C_5H_5)_2} + \mathrm{CH_3COCl} \xrightarrow{\mathrm{AlCl_3}} \mathrm{Fe(C_5H_5)(C_5H_4COCH_3)} + \mathrm{HCl}} \tag{5}$$

The product of this reaction is acetylferrocene, also called monoacetylferrocene. Under conditions in which a second acetyl group is introduced onto the other cyclopentadienyl ring, the overall reaction is:

$$\boxed{\mathrm{Fe(C_5H_5)_2 + 2\,CH_3COCl \xrightarrow{AlCl_3} Fe(C_5H_4COCH_3)_2 + 2\,HCl}} \tag{6}$$

The product of the second reaction is diacetylferrocene. These equations show the one-to-one mole relationship between ferrocene and each isolated product. They also show that formation of diacetylferrocene requires two moles of acetyl chloride for every mole of ferrocene.

The simplified mechanistic sequence is:

$$\mathrm{CH_3COCl + AlCl_3 \longrightarrow CH_3CO^+ + AlCl_4^-} \tag{7}$$

$$\mathrm{Fe(C_5H_5)_2 + CH_3CO^+ \longrightarrow [Fe(C_5H_5)(C_5H_5COCH_3)]^+} \tag{8}$$

$$\mathrm{[Fe(C_5H_5)(C_5H_5COCH_3)]^+ + AlCl_4^- \longrightarrow Fe(C_5H_5)(C_5H_4COCH_3) + AlCl_3 + HCl.} \tag{9}$$

The second step represents electrophilic attack by the acylium ion to give the arenium ion. The final step represents deprotonation of the arenium ion, which restores aromaticity and regenerates aluminum chloride.

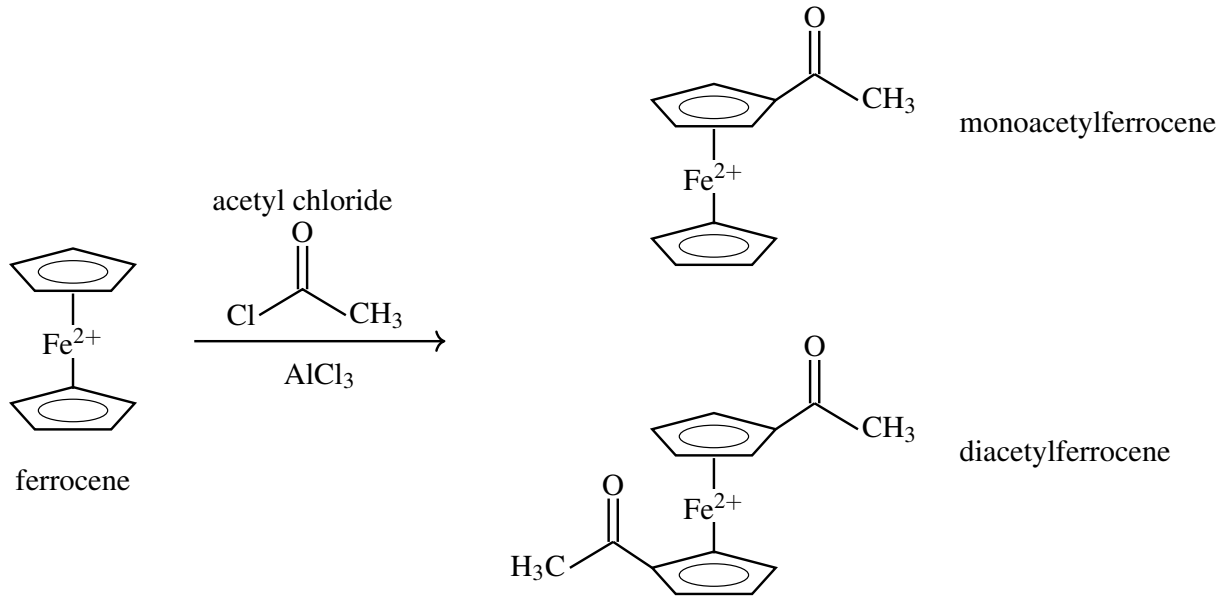


Figure 1: Sandwich-structure reaction scheme for the Friedel–Crafts acylation of ferrocene with acetyl chloride and $AlCl_3$, giving monoacetylferrocene and diacetylferrocene.

2.4 Formation of Diacetylferrocene

The acetyl group is strongly electron-withdrawing. After the first acylation, the substituted ring is deactivated because the carbonyl group withdraws electron density through both inductive and resonance effects. Consequently, a second electrophilic substitution is not favored on the already acylated ring. Under the conditions of this experiment, the second acetyl group can instead be introduced onto the other cyclopentadienyl ring, producing diacetylferrocene.

2.5 Origin of the Acid During Workup

Hydrogen chloride is formed when the arenium ion loses a proton and the proton combines with chloride-containing aluminum species. Additional HCl forms when unreacted acetyl chloride or aluminum chloride reacts with water during the quench. Sodium hydroxide was therefore added gradually to neutralize the acidic reaction mixture:

$$\mathrm{HCl} + \mathrm{NaOH} \longrightarrow \mathrm{NaCl} + \mathrm{H_2O}. \tag{10}$$

2.6 Extraction and Layer Identification

Methylene chloride has a density of approximately $1.33\,\mathrm{g/mL}$, while water has a density of approximately $1.00\,\mathrm{g/mL}$. Therefore, the aqueous layer is the top layer and the methylene chloride organic layer is the bottom layer. The organic layer contains methylene chloride, ferrocene, acetylferrocene, and diacetylferrocene.

2.7 TLC and Column Chromatography Theory

Silica and alumina are polar stationary phases. Ferrocene is the least polar compound in the reaction mixture, so it has the greatest affinity for the relatively nonpolar mobile phase and travels fastest. Acetylferrocene contains one polar carbonyl group and diacetylferrocene contains two; these compounds interact more strongly with the stationary phase and travel more slowly. The same polarity difference explains the column-chromatography sequence: ferrocene elutes with pure hexane, acetylferrocene elutes with 1:1 methylene chloride/hexane, and diacetylferrocene requires the more polar 9:1 methylene chloride/methanol mixture.

$$R_f = \frac{\text{distance traveled by compound}}{\text{distance traveled by solvent front}}. \tag{11}$$

One-spot purity means that a fraction produces only one detectable TLC spot. This indicates that the fraction contains primarily one compound and that the monoacetylferrocene and diacetylferrocene were successfully separated.

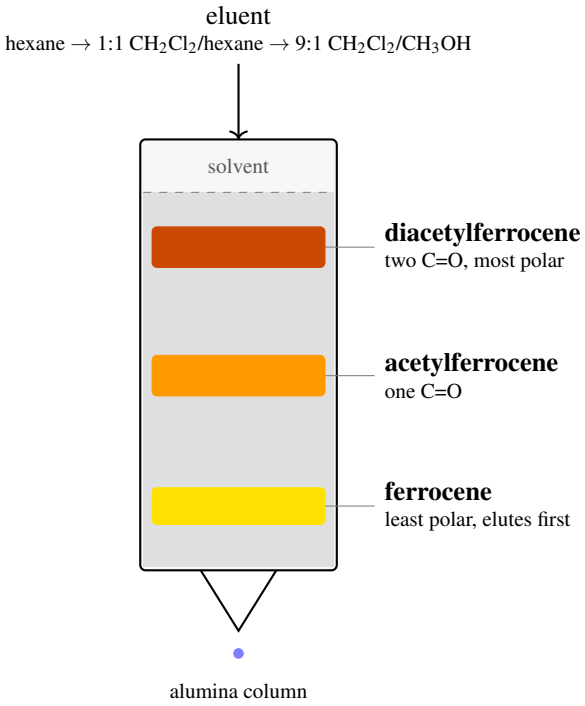


Figure 2: Schematic alumina column during separation. The least-polar ferrocene band moves fastest and exits first, followed by acetylferrocene and then the more-polar diacetylferrocene band.

2.8 Infrared Spectroscopy

Both products contain a ketone functional group, so both should show a strong carbonyl stretching absorption in the approximately 1650–1700 cm^{-1} region. The spectra should also contain sp^2 C–H stretching absorptions near 3100 cm^{-1} and sp^3 C–H stretching absorptions in the approximately 2850–3000 cm^{-1} region. Because acetylferrocene and diacetylferrocene contain the same functional groups, their IR spectra are expected to be similar.

3 Objectives

The objectives of this experiment were to:

1. synthesize acetylferrocene and diacetylferrocene by Friedel–Crafts acylation of ferrocene;
2. use TLC to identify ferrocene, acetylferrocene, and diacetylferrocene in the crude mixture;
3. separate the products by alumina column chromatography;
4. determine the theoretical yield and percent yield of each isolated product;
5. evaluate product purity using TLC and melting point; and
6. identify the major functional-group absorptions in the IR spectra.

4 Hypothesis

If ferrocene is treated with acetyl chloride and anhydrous aluminum chloride, then electrophilic aromatic substitution will produce acetylferrocene and some diacetylferrocene. Ferrocene should have the largest TLC R_f value because it is least polar, acetylferrocene should have an intermediate R_f value, and diacetylferrocene should remain closest to the origin. After column chromatography, purified product fractions should show one TLC spot and IR spectra containing a ketone C=O absorption.

5 Materials and Equipment

5.1 Chemicals

- Ferrocene
- Acetyl chloride
- Anhydrous aluminum chloride, $AlCl_3$
- Methylene chloride, CH_2Cl_2
- Cold water
- Aqueous sodium hydroxide, approximately 6.0 M
- Anhydrous sodium sulfate
- Alumina
- Hexane
- Methanol
- Silica TLC plates and reference solutions

5.2 Equipment

- Tared 5 mL conical vial with spin vane
- Claisen adapter, septum, lid, and drying tube with cotton and blue calcium chloride
- Automatic delivery pipet and fume hood
- Centrifuge tube, litmus paper, and separatory extraction equipment
- Tared beaker, analytical balance, and hot plate
- TLC spotter and developing chamber
- Monster pipet column and alumina
- Melting-point apparatus and Agilent ATR-IR spectrometer

6 Safety Considerations

Safety goggles, appropriate gloves, and a lab coat were worn during the experiment. Acetyl chloride is corrosive, volatile, and reacts with atmospheric moisture to release hydrogen chloride; it was dispensed and capped in the fume hood. Anhydrous aluminum chloride is moisture-sensitive and corrosive. Methylene chloride, hexane, and methanol are volatile organic solvents and were handled in the hood, with hexane and methanol kept away from ignition sources. Sodium hydroxide is strongly caustic and was added slowly to the acidic reaction mixture while the tube was shaken carefully.

The reaction mixture was never heated as a closed system. The drying tube helped exclude moisture during the reaction. Broken glass was disposed of in the designated glass waste, and all organic solvents, alumina, drying agent, and contaminated materials were placed in the appropriate labeled organic waste containers. Skin contact and inhalation of vapors were avoided, and spills or injuries were reported immediately.

7 Methods and Procedure

7.1 Reaction and Workup

A tared 5 mL conical vial equipped with a spin vane was attached to a Claisen adapter. One Claisen head was connected to a drying tube containing cotton and blue calcium chloride, while the other was fitted with a septum and lid. Anhydrous aluminum chloride was added to the vial, followed by methylene chloride. Acetyl chloride was delivered in the hood using an automatic delivery pipet. Ferrocene was dissolved separately in methylene chloride and transferred to the reaction vial. The reaction mixture was stirred at room temperature for 15 min.

The reaction was quenched by transferring the mixture to a centrifuge tube containing 5 mL of cold water. The mixture was neutralized with approximately 6.0 M sodium hydroxide, added a few drops at a time while shaking. The aqueous layer was checked with litmus paper. The organic material was extracted with three 3 mL portions of methylene chloride. The combined organic layers were dried over anhydrous sodium sulfate and transferred to a tared beaker. The solvent was evaporated using low heat in the hood, and the crude product was weighed.

7.2 TLC and Column Chromatography

The crude product was dissolved in approximately 0.5 mL of methylene chloride and mixed with approximately 300 mg of alumina. The mixture was dried and divided into two portions. One portion was loaded onto an alumina column constructed in a Monster pipet. Pure hexane was used first to elute unreacted ferrocene. A 1:1 mixture of methylene chloride and hexane was then used to elute acetylferrocene. Finally, a 9:1 mixture of methylene chloride and methanol was used to elute diacetylferrocene. Fractions were collected separately and analyzed by TLC. Fractions containing the same compound were combined, and solvents were removed using gentle heat in the hood.

7.3 Characterization

The isolated products were weighed, and percent yields were calculated. Melting points were measured and compared with literature values. IR spectra were obtained for the isolated products, and the ketone carbonyl

absorption was identified as the primary diagnostic functional-group peak.

8 Results

8.1 Recorded Experimental Conditions

Table 1: Experimental conditions recorded in the laboratory notebook.

Material or condition	**Recorded amount**	**Notes**
Anhydrous $AlCl_3$	0.152 g	Lewis acid
Methylene chloride, reaction solvent	2.50 mL	CH_2Cl_2
Acetyl chloride	80 µL	Density: 1.11 g/mL
Ferrocene	0.100 g	Dissolved in 1.5 mL CH_2Cl_2
Reaction time	15 min	Room temperature
Quench water	5 mL	Cold water
Sodium hydroxide	6.0 M, 9–10 drops	Aqueous layer checked with litmus
Tared beaker mass	49.551 g	Before solvent removal
Final beaker mass	49.637 g	After solvent removal
Crude product mass	0.086 g	Difference between beaker masses

8.2 TLC Results

The solvent front traveled 6.00 cm. The recorded spot distances were 1.32 cm for acetylferrocene and 0.20 cm for diacetylferrocene. No distance was recorded for ferrocene; its R_f is an estimate based on its position near the solvent front.

Table 2: TLC results using methylene chloride as the mobile phase.

Compound	**Spot distance (cm)**	R_f
Ferrocene	not recorded (estimate)	≈ 0.90 (estimate)
Acetylferrocene	1.32	0.220
Diacetylferrocene	0.20	0.0333

$$R_f = \frac{\text{distance traveled by compound}}{\text{distance traveled by solvent front}} \tag{12}$$

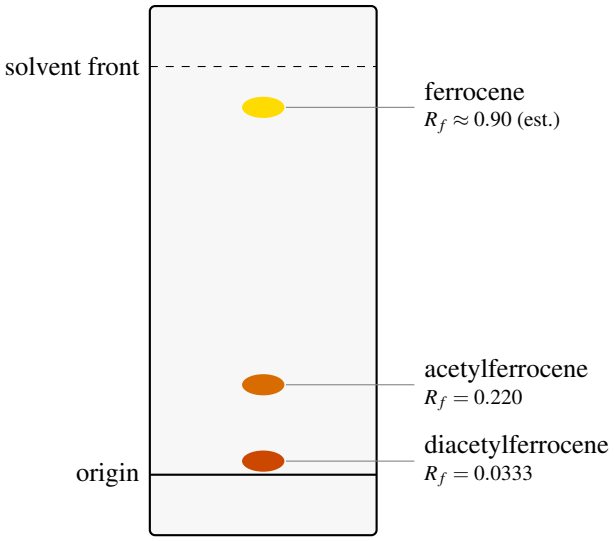


Figure 3: Schematic TLC plate (silica, CH_2Cl_2) drawn to scale from the recorded R_f values; the solvent front is 6.00 cm. The ferrocene position is an estimate because no distance was recorded.

8.3 Isolated Products

Table 3: Masses and physical characterization of isolated products.

Compound	Molar mass (g/mol)	Mass (g)	Literature m.p.	Experimental m.p.
Acetylferrocene	228.07	0.120	81–83 °C	85.5 °C
Diacetylferrocene	270.11	0.065	125–127 °C	112 °C

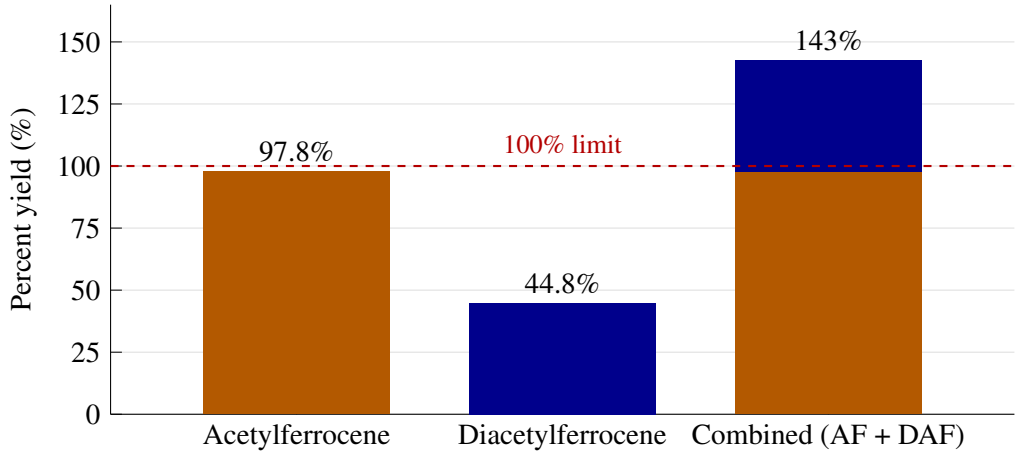


Figure 4: Percent yields of the isolated products, each calculated from 0.100 g of ferrocene. The combined bar stacks the two yields (orange = acetylferrocene, blue = diacetylferrocene) and exceeds the 100 % limit, showing that the recorded masses are inconsistent (see Discussion).

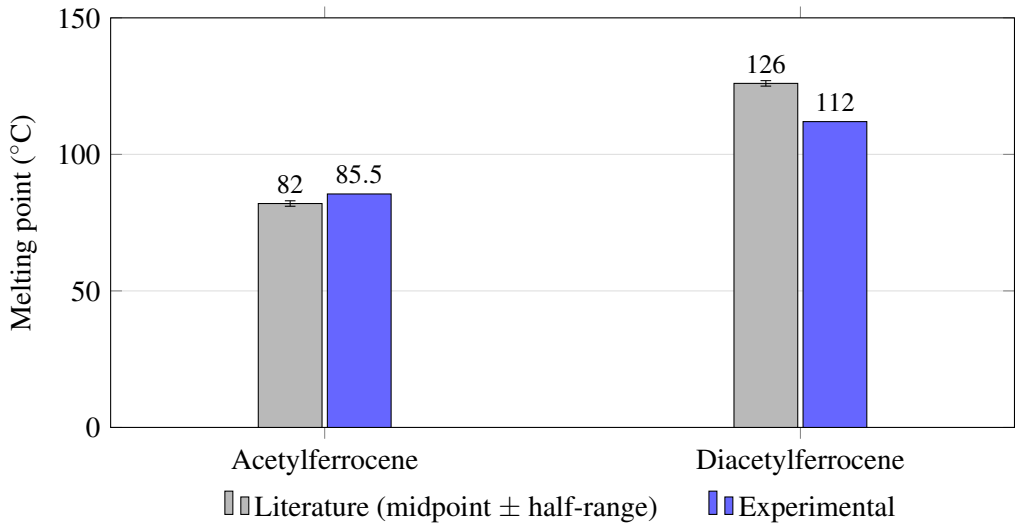


Figure 5: Experimental melting points compared with the midpoint of each literature range (error bars show the literature range).

8.4 IR Spectroscopy

The notebook contains hand-drawn IR traces rather than exported numerical spectra. The important observed features were a strong ketone carbonyl absorption near the 1650–1700 cm^{-1} region and C–H stretching absorptions in the approximately 2850–3100 cm^{-1} region. The acetylferrocene and diacetylferrocene spectra were similar because both compounds contain ketone carbonyl groups and ferrocene rings.

Table 4: Diagnostic IR absorptions identified from the recorded spectra.

Assignment	**Approximate region (cm^{-1})**	**Interpretation**
Ketone C=O stretch	1650–1700	Confirms an acyl ketone group
sp^2 C–H stretch	approximately 3000–3100	Ferrocene-ring C–H bonds
sp^3 C–H stretch	approximately 2850–3000	Methyl-group C–H bonds

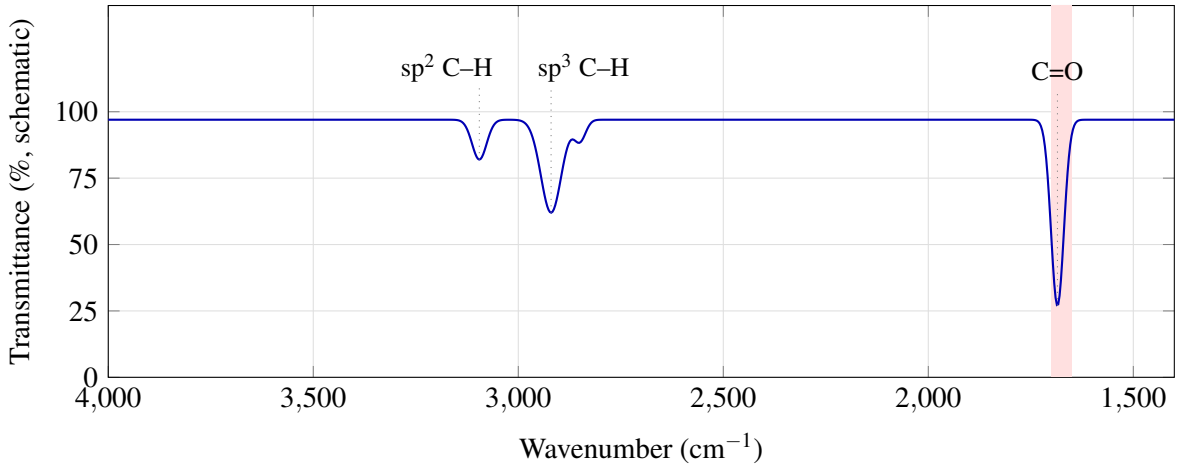


Figure 6: Schematic IR profile reconstructed from the hand-drawn notebook spectrum (peaks point downward, as in a transmittance spectrum). The shaded band marks the ketone C=O region, 1650–1700 cm^{-1}. This figure is not a digitized instrument output.

9 Calculations

9.1 Crude Product Mass

$$\begin{aligned} m_{\text{crude}} &= m_{\text{final beaker}} - m_{\text{tared beaker}} \\ &= 49.637\,\text{g} - 49.551\,\text{g} = \boxed{0.086\,\text{g}} \end{aligned}$$

9.2 Moles of Starting Materials

For ferrocene, $M = 186.03\,\text{g/mol}$:

$$\begin{aligned} n_{\text{ferrocene}} &= \frac{0.100\,\text{g}}{186.03\,\text{g/mol}} \\ &= \boxed{5.38 \times 10^{-4}\ \text{mol}} \end{aligned}$$

The mass of acetyl chloride was calculated from its volume and density:

$$\begin{aligned} m_{\text{acetyl chloride}} &= 80\,\mu\text{L}\left(\frac{1\,\text{mL}}{1000\,\mu\text{L}}\right)\left(\frac{1.11\,\text{g}}{1\,\text{mL}}\right) \\ &= 0.0888\,\text{g} \end{aligned}$$

Using $M_{\text{acetyl chloride}} = 78.50\,\text{g/mol}$:

$$n_{\text{acetyl chloride}} = \frac{0.0888\,\text{g}}{78.50\,\text{g/mol}}$$
$$= \boxed{1.13 \times 10^{-3}\ \text{mol}}$$

9.3 Limiting Reagent and Theoretical Yields

Formation of acetylferrocene requires one mole of acetyl chloride per mole of ferrocene:

$$\text{ferrocene} + \text{CH}_3\text{COCl} \longrightarrow \text{acetylferrocene} + \text{HCl} \tag{13}$$

The ferrocene-based theoretical mass is:

$$m_{\text{theoretical, AF}} = 5.38 \times 10^{-4}\ \text{mol}\left(\frac{228.07\,\text{g}}{\text{mol}}\right)$$
$$= \boxed{0.1227\,\text{g}}$$

The acetyl chloride-based theoretical mass is:

$$m_{\text{from AcCl, AF}} = 1.13 \times 10^{-3}\ \text{mol}\left(\frac{228.07\,\text{g}}{\text{mol}}\right)$$
$$= 0.258\,\text{g}$$

Because ferrocene gives the smaller theoretical amount, ferrocene is the limiting reagent for acetylferrocene.

Formation of diacetylferrocene requires two moles of acetyl chloride per mole of ferrocene:

$$\text{ferrocene} + 2\,\text{CH}_3\text{COCl} \longrightarrow \text{diacetylferrocene} + 2\,\text{HCl} \tag{14}$$

From ferrocene:

$$m_{\text{theoretical, DAF}} = 5.38 \times 10^{-4}\ \text{mol}\left(\frac{270.11\,\text{g}}{\text{mol}}\right)$$
$$= \boxed{0.145\,\text{g}}$$

From acetyl chloride:

$$m_{\text{from AcCl, DAF}} = \frac{1.13 \times 10^{-3}}{2}\ \text{mol}\left(\frac{270.11\,\text{g}}{\text{mol}}\right)$$
$$= 0.153\,\text{g}$$

Ferrocene is also the limiting reagent for diacetylferrocene because 0.145 g is less than 0.153 g. The 80 µL of acetyl chloride is sufficient to provide the two acetyl groups required per mole of ferrocene.

9.4 Percent Yield

For acetylferrocene:

$$\begin{aligned}\%\text{yield}_{\text{AF}} &= \frac{0.120\,\text{g}}{0.1227\,\text{g}} \times 100 \\ &= \boxed{97.8\,\%}\end{aligned}$$

For diacetylferrocene:

$$\begin{aligned}\%\text{yield}_{\text{DAF}} &= \frac{0.065\,\text{g}}{0.145\,\text{g}} \times 100 \\ &= \boxed{44.8\,\%}\end{aligned}$$

The notebook recorded 95.24 % for acetylferrocene and 50.38 % for diacetylferrocene. The values reported here use the balanced reaction stoichiometry and the molar mass of diacetylferrocene, $C_{14}H_{14}FeO_2$, which is $270.11\,\text{g/mol}$.

9.5 Mass Balance Check

Because both products form from the same ferrocene, the moles of ferrocene consumed must not exceed the moles initially present:

$$\begin{aligned}n_{\text{AF}} &= \frac{0.120\,\text{g}}{228.07\,\text{g/mol}} = 5.26 \times 10^{-4}\ \text{mol} \\ n_{\text{DAF}} &= \frac{0.065\,\text{g}}{270.11\,\text{g/mol}} = 2.41 \times 10^{-4}\ \text{mol} \\ \frac{n_{\text{AF}} + n_{\text{DAF}}}{n_{\text{ferrocene}}} &= \frac{7.67 \times 10^{-4}}{5.38 \times 10^{-4}} \times 100 = \boxed{143\,\%}\end{aligned}$$

A combined value above 100 % is physically impossible. Acetyl chloride is not the problem: the products contain $5.26 \times 10^{-4} + 2(2.41 \times 10^{-4}) = 1.01 \times 10^{-3}$ mol of acetyl groups, which is less than the 1.13×10^{-3} mol supplied.

9.6 Melting-Point Comparison

The percent difference from the midpoint of the literature melting range was calculated as:

$$\%\text{difference} = \frac{|T_{\text{experimental}} - T_{\text{literature midpoint}}|}{T_{\text{literature midpoint}}} \times 100 \tag{15}$$

For acetylferrocene, the literature midpoint is 82 °C:

$$\%\text{difference}_{\text{AF}} = \frac{|85.5 - 82|}{82} \times 100 = \boxed{4.27\%}$$

For diacetylferrocene, the literature midpoint is 126 °C:

$$\%\text{difference}_{\text{DAF}} = \frac{|112 - 126|}{126} \times 100 = \boxed{11.1\%}$$

10 Discussion

The reaction produced both monoacetylferrocene and diacetylferrocene, as expected for Friedel–Crafts acylation of ferrocene under the experimental conditions. The TLC behavior supported the expected polarity order. Ferrocene traveled farthest because it interacted least strongly with silica. Acetylferrocene traveled less far because its ketone group increased polarity, while diacetylferrocene remained closest to the origin because it contained two ketone groups and interacted most strongly with the polar stationary phase.

Yield and mass-balance inconsistency. The isolated masses are not mutually consistent. Individually, 0.120 g of acetylferrocene is 97.8 % of the ferrocene-based theoretical yield and 0.065 g of diacetylferrocene is 44.8 %. But each mole of product consumes one mole of ferrocene, so the two yields must sum to no more than 100 % (and less if any ferrocene was recovered unreacted). The recorded masses correspond to 143 % of the ferrocene used. In addition, the crude mass after workup was only 0.086 g, less than the 0.185 g of purified products reported after the column. A purified product cannot outweigh the crude mixture it came from, so the reported masses are almost certainly too high.

The most likely explanations are (i) residual solvent, particularly methylene chloride/methanol held in the diacetylferrocene fraction and hexane or methylene chloride in the acetylferrocene fraction, since the fractions were dried only with gentle heat; (ii) alumina or other solids carried over into the collected fractions; (iii) a tared-mass or balance-reading error for the crude product, which would make the crude mass artificially low; and (iv) incomplete drying of the crude beaker so that the crude mass was recorded inconsistently. Because the same 0.100 g of ferrocene cannot support both reported masses, the individual percent yields should be regarded as upper bounds rather than reliable measures of reaction efficiency. Reweighing the dried products would resolve the question. A more defensible conclusion from these data is only that both products were formed and isolated, not that acetylferrocene was obtained in near-quantitative yield.

If instead the crude mass of 0.086 g is taken as correct, the maximum possible product is well below 0.185 g, which reinforces the conclusion that the product masses include non-product material. The crude mass is also lower than the starting ferrocene mass, which is plausible only because aqueous workup, extraction, drying, and transfers lose material; this is a further indication that the isolated masses, rather than the crude mass, are inflated.

The experimental melting point of acetylferrocene, 85.5 °C, was slightly above the literature range of 81–83 °C. A melting point above the literature range is unusual for an impure sample, so this may reflect thermometer calibration or heating rate rather than purity. The experimental melting point of diacetylferrocene, 112 °C, was substantially below the literature range of 125–127 °C. A depressed melting point generally indicates contamination, incomplete drying, incomplete separation, or an inaccurate measurement. This is consistent with the mass-balance problem above: residual solvent or contamination by acetylferrocene would both raise the apparent mass and lower the melting point.

Both IR spectra showed a strong absorption in the ketone carbonyl region near 1650–1700 cm^{-1}. The spectra also contained C–H stretching absorptions in the approximately 2850–3100 cm^{-1} region. These features support the presence of acetyl groups in both isolated products. Because the products have the same functional groups, their IR spectra are expected to be similar and IR alone cannot distinguish them; TLC R_f values and melting points are needed for that.

11 Conclusion

Ferrocene was subjected to Friedel–Crafts acylation using acetyl chloride and anhydrous aluminum chloride. TLC showed the expected separation behavior: ferrocene traveled fastest, acetylferrocene traveled more slowly, and diacetylferrocene remained closest to the origin. Alumina column chromatography yielded fractions identified as acetylferrocene (0.120 g) and diacetylferrocene (0.065 g). Calculated individually from 0.100 g of ferrocene, the yields were 97.8 % and 44.8 %, but together they correspond to 143 % of the ferrocene used, so the isolated masses are overestimated, most likely because of residual solvent or impurities. The acetylferrocene melting point was close to the literature value, whereas the low diacetylferrocene melting point indicated an impure or incompletely dried fraction. The IR carbonyl absorptions supported the presence of the expected ketone functional group in both products.

Waste Disposal

Organic liquid waste (methylene chloride, hexane, methanol), alumina, sodium sulfate, and contaminated solids were placed in the appropriate labeled waste containers. Aqueous waste was neutralized before disposal as directed by the instructor.

References

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3. Silverstein, R. M.; Webster, F. X.; Kiemle, D. J.; Bryce, D. L. *Spectrometric Identification of Organic Compounds*, 8th ed.; Wiley: Hoboken, NJ, 2015.